

Overview and update on methods for cargo loading into extracellular vesicles

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Abstract

The enormous library of pharmaceutical compounds presents endless research avenues. However, several factors limit the therapeutic potential of these drugs, such as drug resistance, stability, off-target toxicity, and inadequate delivery to the site of action. Extracellular vesicles (EVs) are lipid bilayer-delimited particles and are naturally released from cells. Growing evidence shows that EVs have great potential to serve as effective drug carriers. Since EVs can not only transfer biological information, but also effectively deliver hydrophobic drugs into cells, the application of EVs as a novel drug delivery system has attracted considerable scientific interest. Recently, EVs loaded with siRNA, miRNA, mRNA, CRISPR/Cas9, proteins, or therapeutic drugs show improved delivery efficiency and drug effect. In this review, we summarize the methods used for the cargo loading into EVs, including siRNA, miRNA, mRNA, CRISPR/Cas9, proteins, and therapeutic drugs. Furthermore, we also include the recent advance in engineered EVs for drug delivery. Finally, both advantages and challenges of EVs as a new drug delivery system are discussed. Here, we encourage researchers to further develop convenient and reliable loading methods for the potential clinical applications of EVs as drug carriers in the future.